Chapter 6: The Theory of the Firm: Production and Cost Curves

Principles of Microeconomics Curriculum

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Abstract

This chapter provides a rigorous, pedagogical treatment of Chapter 6: The Theory of the Firm: Production and Cost Curves, benchmarked to standard principles curriculum.

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1 Production Functions and Diminishing Returns

Output $Q = f(L, \bar{K})$. The Marginal Product of Labor is $MP_L = \frac{\Delta Q}{\Delta L}$. By the Law of Diminishing Marginal Productivity, $\frac{d(MP_L)}{dL} < 0$.

2 Cost Duality and Minimum Crossings

Total Cost $TC = FC + VC(Q)$. Marginal Cost is inversely linked to productivity: $MC = \frac{w}{MP_L}$. Average Total Cost $ATC = AFC + AVC$. MC intersects ATC and AVC at their minimum points:

$$\frac{d(ATC)}{dQ} = \frac{MC - ATC}{Q} = 0 \implies MC = ATC \tag{1}$$